GRAINS FUTURES STRATEGY

Complete mechanics, feature calculations, portfolio construction, PnL, regimes, and trade instruments

Outright long/short cross-sectional relative-value strategy over CORN, SOYBEAN, WHEAT_SRW, and WHEAT_HRW.

```
Core idea:
Long grains with strongest positive expected PnL
Short grains with weakest or most negative expected PnL
Mostly relative-value, not a pure directional grain bet
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Prepared as an interview-ready technical memo. All formulas are written in implementation terms so they can be translated directly into Python.

Executive summary

Main forces that move grain futures: weather shocks, crop supply and demand, inventories, export demand, planting and harvest cycles, speculative positioning, commercial hedging, and nearby physical tightness.

The strategy is primarily a cross-sectional relative-value model. It ranks grains by expected risk-adjusted PnL and trades the strongest versus the weakest. The model does not mainly ask, 'Are all grains bullish?' It asks, 'Which grain should outperform the other grains?'

Component	Final choice	Reason
Core trade	Outright futures long/short	Best signal and best tradable PnL in the supplied data.
Main model	Static edge-filtered two-block Ridge	Best candidate performance and simple economic logic.
Robustness check	Annual 20-day walk-forward Ridge	Checks that the result survives retraining using only past data.
Useful extension	80% outright + 20% inter-commodity spread overlay	Improves diversification and drawdown, but gives up some OOS PnL.
Rejected	Broad one-day Ridge, pure ML, calendar-spread-only, hard-coded trade-war rule	Too noisy, too overfit, too weak, or too event-specific.

Important clarification
The final Ridge implementation uses standardized or z-scored feature inputs. Tanh transformations are useful for hand-built signal sleeves, but the main Ridge model does not need tanh because Ridge learns coefficients on standardized features. This PDF still shows both: the final Ridge pipeline and the optional z-score/tanh style used for explainable signal engineering.

1. Detailed pseudo-code: full strategy mechanics

```
INPUTS
- Daily adjusted front futures prices for CORN, SOYBEAN, WHEAT_SRW, WHEAT_HRW
- Second futures price for curve features
- COT positioning data
- Public inventory and receipt data
- Cargill internal inventory and crush data
- Contract size, cost assumptions, and split date

FOR each date t and commodity i:

1. Build price features
mom_60_i,t = price_i,t / price_i,t-60 - 1
rev_5_i,t = - (price_i,t / price_i,t-5 - 1)
vol_20_i,t = std(daily_return_i over last 20 days) * sqrt(252)

2. Build curve features
curve_spread_i,t = C1_i,t - C2_i,t
curve_ratio_i,t = C1_i,t / C2_i,t
curve_change_i,t = curve_spread_i,t - curve_spread_i,t-20

3. Build COT features
cot_mm_level_i,t = money_manager_net_i,t / open_interest_i,t
cot_pm_oi_level_i,t = producer_merchant_net_i,t / open_interest_i,t

4. Build physical and Cargill features
public_inventory_change_i,t = inventory_i,t - inventory_i,t-1
receipts_change_i,t = receipts_i,t - receipts_i,t-1
cgl_inventory_change_i,t = cgl_inventory_i,t - cgl_inventory_i,t-1
crush_surprise_i,t = actual_crush_i,t - expected_or_planned_crush_i,t
crush_utilization_i,t = actual_crush_i,t / capacity_or_planned_crush_i,t

5. Standardize features using training information only
z_i,t = (x_i,t - rolling_or_train_mean_i) / rolling_or_train_std_i

6. Build target
forward_pnl_5_i,t = contract_size * (price_i,t+5 - price_i,t)

7. Fit two Ridge models
core_model: price + curve + COT features, alpha approx 25
physical_model: public physical + Cargill features, alpha approx 1000

8. Predict expected PnL
pred_i,t = core_pred_i,t + physical_pred_i,t

9. Risk-adjust prediction
recent_pnl_vol_i,t = rolling_std(one_day_pnl_i, lookback)
risk_adj_pred_i,t = pred_i,t / recent_pnl_vol_i,t
```

```
10. Cross-sectional relative-value signal
signal_i,t = risk_adj_pred_i,t - mean(risk_adj_pred_all_commodities,t)

11. Scale to fixed gross lot exposure
gross_raw_t = sum(abs(signal_i,t))
if gross_raw_t > 0:
position_i,t = signal_i,t / gross_raw_t
else:
position_i,t = 0

12. Edge filter
edge_t = std(risk_adj_pred_all_commodities,t)
threshold_t = expanding_median(edge up to t-1)
if edge_t <= threshold_t:
position_i,t = 0 for all i

13. Backtest with lagged execution
gross_pnl_t = sum_i(position_i,t-1 * one_day_pnl_i,t)
turnover_t = sum_i(abs(position_i,t - position_i,t-1))
trade_cost_t = turnover_t * cost_per_lot
net_pnl_t = gross_pnl_t - trade_cost_t - holding_cost_t
equity_t = equity_t-1 + net_pnl_t
```

2. Feature normalization: z-score, tanh, and why they matter

The final Ridge pipeline is naturally compatible with z-scored inputs. This makes features comparable and makes Ridge coefficients easier to interpret. A one-unit coefficient then approximately means one standard deviation of that feature. Tanh is useful when constructing hand-built signals because it compresses extreme z-scores and prevents one outlier from dominating the signal.

Transformation	Formula	Use	Interpretation
Return	ret_t = price_t / price_t-1 - 1	Base input for momentum, reversal, volatility, and PnL.	Positive return means price rose.
Rolling z-score	z_t = (x_t - mean(x, window)) / std(x, window)	Makes different signals comparable.	z = +2 means two standard deviations above normal.
Expanding lagged z-score	z_t = (x_t - mean(x up to t-1)) / std(x up to t-1)	Avoids lookahead in live style calculations.	Only uses information known before date t.
Tanh squashing	sig_t = tanh(z_t / 2)	Optional for hand-built signal sleeves.	Maps large z-scores into roughly -1 to +1.
Signed reversal	sig_t = tanh(-z_t / 2)	Mean-reversion signals.	If z is high, signal turns bearish.
Signed momentum	sig_t = tanh(z_t / 2)	Trend-following signals.	If z is high, signal turns bullish.

For the final Ridge model, the essential step is z-scoring or standardizing the features. Tanh is not required by Ridge, but it is useful when explaining directional signal intuition or building rule-based sleeves.

3. Feature calculations and economic rationale

3.1 Price features

Signal	Meaning	Calculation	Trading implication	Why it can work / fail
mom_60	60-day momentum; medium-term trend over about three months.	mom_60 = price_t / price_t-60 - 1. Optional normalized signal: tanh(zscore(mom_60)/2).	Positive = commodity has been trending up; potentially bullish versus peers.	Works in persistent supply or demand shocks. Fails in choppy/sideways markets.
rev_5	5-day reversal; short-term overreaction.	rev_5 = - (price_t / price_t-5 - 1). Optional signal: tanh(zscore(rev_5)/2).	If price rallied too much recently, reversal signal becomes bearish.	Works when weather headlines or positioning overshoot. Fails in real breakout trends.
vol_20	20-day realized volatility; recent instability.	vol_20 = std(daily_returns over 20 days) * sqrt(252).	High volatility means the same prediction should receive smaller risk weight.	Useful for sizing. It is not necessarily alpha by itself.

The strategy learned that one-day prediction is too noisy. A 5-day forward PnL target is more stable for the static candidate, while the robustness check uses a slower 20-day target.

3.2 Curve features

Signal	Meaning	Calculation	Trading implication	Why it can work / fail
curve_spread	Front minus second contract; proxy for backwardation or contango.	curve_spread = C1 - C2.	Positive = front stronger than second; nearby tightness; often bullish.	Works because term structure reflects scarcity. Fails if roll adjustment or delivery effects distort the signal.

Signal	Meaning	Calculation	Trading implication	Why it can work / fail
curve_ratio	Normalized curve tightness.	curve_ratio = C1 / C2.	> 1 = front contract stronger than second contract.	Useful across different price levels. Can be noisy around contract rolls.
curve_change	Whether tightness is increasing or decreasing.	curve_change_20 = curve_spread_t - curve_spread_t-20.	Rising curve = nearby market tightening; potentially bullish.	Works in repricing. Fails if curve moves are technical rather than fundamental.

Economic interpretation: backwardation usually means nearby scarcity, low inventories, or strong immediate demand. Contango usually means abundant supply, weaker immediate demand, or storage pressure.

3.3 COT positioning features

Signal	Meaning	Calculation	Trading implication	Why it can work / fail
cot_mm_level	Money-manager net positioning; proxy for speculative flow and crowding.	cot_mm_level = money_manager_net / open_interest.	Very long specs can mean trend confirmation or crowded reversal risk.	Works because specs often follow trends. Fails when crowded positioning remains crowded or policy headlines dominate.
cot_pm_oi_level	Producer/merchant position normalized by open interest; commercial hedging pressure.	cot_pm_oi_level = producer_merchant_net / open_interest.	Heavy producer selling can imply supply pressure; commercial extremes may contain physical information.	Works because commercials are close to physical flows. Fails because COT is weekly and slow.

3.4 Public physical features

Signal	Meaning	Calculation	Trading implication	Why it can work / fail
public_inventory_change	Change in publicly observed inventories.	inventory_change = inventory_t - inventory_t-1. Optional surprise: -(change - seasonal_mean) / seasonal_std.	Inventory build = looser supply, bearish. Draw = tighter supply, bullish.	Works when inventories are binding. Fails when data is local, lagged, or macro/policy dominates.
receipts_change	Change in grain receipts into elevators, terminals, or plants.	receipts_change = receipts_t - receipts_t-1.	Higher receipts = more supply arriving; usually bearish. Lower receipts = constrained supply; bullish.	Works as a flow proxy. Fails if receipts reflect logistics timing rather than true demand/supply.

3.5 Cargill internal features

Signal	Meaning	Calculation	Trading implication	Why it can work / fail
cgl_inventory_change	Change in Cargill internal inventory.	cgl_inventory_change = cgl_inv_t - cgl_inv_t-1.	Inventory build = potentially bearish; draw = potentially bullish.	Can be faster/cleaner than public data. Fails if internal footprint is not representative.
crush_surprise	Actual crush activity versus planned or expected crush.	crush_surprise = actual_crush - expected_crush, or processed - planned.	Positive surprise = stronger processing demand, especially for soybeans.	Useful for soybean demand. Fails for non-soy grains or when crush data is aggregate.
crush_utilization	How much crush capacity is being used.	crush_utilization = actual_crush / capacity, or processed / planned.	High utilization = strong processing demand.	Works if utilization reflects real demand. Fails if planned volumes are revised or capacity estimates are noisy.

4. Model structure: two-block Ridge

The model separates stable core signals from noisier physical signals. This is one of the most important design decisions.

Block	Inputs	Target	Model	Why
Core model	mom_60, rev_5, curve_spread, curve_ratio, cot_mm_level, cot_pm_oi_level	5-day forward one-lot PnL	Ridge regression, alpha around 25	Price, curve, and COT are the most stable signal families.
Physical overlay	public_inventory_change, receipts_change, cgl_inventory_change, crush_surprise, crush_utilization	5-day forward one-lot PnL	Ridge regression, alpha around 1000	Physical data can help, but should be heavily regularized because it is sparse/noisy.

```
Core prediction:
pred_core_i,t = Ridge_core(z_core_features_i,t)

Physical overlay prediction:
pred_phys_i,t = Ridge_phys(z_physical_features_i,t)

Final prediction:
pred_i,t = pred_core_i,t + overlay_weight * pred_phys_i,t

Selected overlay weight in the static candidate: full physical overlay weight.
Reason: it improved validation/full-period behavior while keeping Cargill data controlled by strong
```

```
regularization.
```

Why Ridge? The feature set is correlated: curve spread and curve ratio are related; positioning variables can also move together. Ridge shrinks coefficients and reduces overfit compared with ordinary least squares on a small sample.

5. Target, futures PnL, and unit definitions

The model predicts one-lot dollar PnL, not just price direction. For grain futures, one full-size contract is commonly represented as 5,000 bushels in this research setup.

```
One-day one-lot PnL:
one_day_pnl_i,t = contract_size * (price_i,t - price_i,t-1)

5-day forward target used by static model:
target_5_i,t = contract_size * (price_i,t+5 - price_i,t)

20-day forward target used by annual walk-forward robustness model:
target_20_i,t = contract_size * (price_i,t+20 - price_i,t)

Lag rule:
A target ending at t+H cannot be used for training until date t+H is known.
This avoids lookahead leakage.
```

Answer to your question
Yes, the complete mechanics include gross PnL, net PnL, turnover, trade costs, holding costs, position scaling, z-score normalization, and optional tanh signal compression. The final implementation uses Ridge on standardized features; tanh is optional for rule-based signal sleeves, not necessary for the main Ridge model.

6. Position construction: raw signals to gross exposure

The strategy turns predictions into a relative-value book. The strongest predicted grains receive positive positions, and the weakest predicted grains receive negative positions.

```
Step 1: risk-adjust predictions
risk_adj_pred_i,t = pred_i,t / recent_pnl_vol_i,t

Step 2: cross-sectional demeaning
avg_t = mean(risk_adj_pred_i,t across commodities)
raw_signal_i,t = risk_adj_pred_i,t - avg_t

Step 3: scale to one aggregate gross lot
gross_raw_t = sum_i(abs(raw_signal_i,t))
position_i,t = raw_signal_i,t / gross_raw_t

Properties:
sum_i(position_i,t) approximately 0 because of cross-sectional demeaning
sum_i(abs(position_i,t)) = 1 if the book is active

Example:
risk_adj_pred = [1.2, 0.4, -0.2, -1.0]
mean = 0.1
raw_signal = [1.1, 0.3, -0.3, -1.1]
gross_raw = 2.8
positions = [0.393, 0.107, -0.107, -0.393]
```

Interpretation: the book is mostly market-neutral. It profits when the long grains outperform the short grains, rather than from all grains going up together.

7. Edge filter and opportunity filters

The most important risk-control idea is to trade only when the model has a clear cross-sectional opinion. If all risk-adjusted predictions are close together, the model has no strong relative-value edge.

```
Prediction-dispersion edge filter:
edge_t = std(risk_adj_pred_i,t across commodities)
threshold_t = expanding_median(edge_s for s < t)

if edge_t > threshold_t:
keep positions
else:
set all positions to zero
```

Filter	Definition	Use	Comment
Prediction dispersion	std of risk-adjusted predictions	Final static edge-filtered candidate	Avoids low-conviction days.

Filter	Definition	Use	Comment
Curve dispersion	cross-sectional dispersion of curve signals	Tested in multi-condition filter	Trades more when curve differences are meaningful.
Momentum dispersion	cross-sectional dispersion of momentum signals	Tested in multi-condition filter	Trades more when price trends separate across grains.
Multi-condition filter	prediction, curve, and momentum dispersion all above expanding thresholds	High-conviction extension	Improves some weak regimes but reduces total PnL because it trades fewer days.

8. Backtest accounting: gross PnL, net PnL, costs, and equity

Backtesting must use lagged positions. A position decided at date t earns PnL from the next price move, not the same-day move used to create the signal.

```
For each day t:

1. Lag positions
trade_position_i,t = position_i,t-1

2. Gross futures PnL
gross_pnl_t = sum_i(trade_position_i,t * one_day_pnl_i,t)

3. Turnover in lots
turnover_t = sum_i(abs(position_i,t - position_i,t-1))

4. Trade cost
trade_cost_t = turnover_t * cost_per_lot

5. Optional margin holding/funding cost
margin_used_t = sum_i(abs(position_i,t-1) * margin_per_contract_i)
holding_cost_t = margin_used_t * annual_funding_rate / 252

6. Net PnL
net_pnl_t = gross_pnl_t - trade_cost_t - holding_cost_t

7. Cumulative equity
equity_t = equity_t-1 + net_pnl_t

8. Drawdown
running_peak_t = max(equity_s for s <= t)
drawdown_t = equity_t - running_peak_t
```

Metric	Formula	Meaning
Gross exposure	sum(abs(position_i,t))	Total absolute lots traded. Final active book is usually scaled to about one gross lot.
Net exposure	sum(position_i,t)	Directional bias. Cross-sectional demeaning keeps this near zero.
Gross PnL	sum(position_t-1 * one_day_pnl_t)	PnL before costs.
Turnover	sum(abs(position_t - position_t-1))	How much the book changed.
Trade cost	turnover * cost_per_lot	Transaction cost from changing positions.
Net PnL	gross_pnl - trade_cost - holding_cost	True tradable PnL after implementation costs.
Sharpe	mean(daily_net_pnl) / std(daily_net_pnl) * sqrt(252)	Risk-adjusted return.
Max drawdown	min(equity - running_peak)	Worst peak-to-trough loss.

9. Holding period, smoothing, and risk controls

The final strategy uses daily rebalancing even though it predicts multi-day PnL. Holding-period experiments were run to check if turnover could be reduced without losing edge.

Mechanic	Formula / implementation	Result
Daily rebalance	Update positions every trading day.	Final default for robustness model.
Skip-rebalance 2d	Update only every second trading day; hold flat between updates.	Best static execution candidate; improves static OOS Sharpe and lowers turnover.
Staggered hold	Average N offset sleeves refreshed on different days.	Smoother but generally weaker because it dilutes conviction.
EWMA smoothing	smoothed_t = alpha * signal_t + (1-alpha) * smoothed_t-1, alpha = 1 - exp(-ln(2)/halflife).	Generally did not improve final Ridge results.

Mechanic	Formula / implementation	Result
Vol targeting	vol_scale = target_vol / realized_vol, clipped by leverage cap.	Tested, but did not beat the annual Ridge benchmark.
Crisis cap	cap exposure when realized vol > 2x long-run vol.	Useful conceptually, but not promoted in final grains strategy.

10. Performance summary and model selection

Strategy	OOS Sharpe	OOS PnL	Full Sharpe	Max DD	Decision
Old broad one-day Ridge	-0.460	-4,193	0.711	n/a	Reject: clear overfit.
60-day momentum baseline	0.619	5,866	-0.313	n/a	Benchmark only.
Static unfiltered two-block Ridge	1.087	9,461	0.673	-7,449	Useful but weaker risk control.
Static edge-filtered two-block Ridge	1.498	10,304	1.077	-4,648	Main tradable candidate.
Annual 20-day walk-forward Ridge	1.347	11,579	0.738	-4,001	Main anti-overfit check.
Static edge-filtered, 2-day skip rebalance	1.730	11,637	1.087	-4,111	Best static execution candidate; needs holdout validation.
Best ML: shallow gradient boosting	1.274	10,450	0.633	-3,164	Close but worse and higher overfit risk.

The static edge-filtered model is the best candidate, but the annual walk-forward model is the cleaner proof that the idea is not purely a static-coefficient artifact.

11. Regime behavior

Regime	Behavior	Reason	Risk / response
Drought / weather shock	Good	Trend, curve tightness, and positioning often align.	Use model normally; expect high dispersion.
Export ban / geopolitics	Good	Nearby scarcity and repricing create cross-sectional differences.	Outrights work well; monitor headline risk.
COVID recovery / China buying	Very good	Demand repricing and strong cross-sectional flows.	Model can capture relative winners.
Low-price abundant supply	Weak	Low dispersion and weaker trend/curve signals.	Edge filter should reduce activity.
US-China trade war	Weak / negative	Policy and tariff headlines dominate normal fundamentals, especially soybeans.	Do not hard-code tariff rule; use as documented regime risk.
Choppy sideways market	Weak	Price and curve signals become noisy.	Trade less; use edge/opportunity filters.

Key point: this is a shock/repricing and relative-value strategy. It works best when one grain market becomes tighter or looser than the others. It is not a stable carry strategy that should be expected to earn evenly in all environments.

12. What exactly do we trade? Outrights, spreads, and crush

12.1 Core book: outright futures

The core book trades long/short outright futures in the four grain contracts. The final position is determined by risk-adjusted expected PnL and cross-sectional scaling.

```
Example outright book:
Long CORN
Long SOYBEAN
Short WHEAT_SRW
Short WHEAT_HRW

Interpretation:
The model expects corn and soybeans to outperform the wheat contracts over the forecast horizon.
```

12.2 Calendar spreads

Calendar spreads were tested as tradable instruments, not just as predictors. The result was too weak to promote.

```
Calendar spread unit:
Long front contract C1
Short second contract C2

Calendar spread PnL:
calendar_spread_pnl_t = contract_size * delta(C1_t - C2_t)
```

```
Decision:
Rejected as final instrument. Calendar spreads likely need delivery-month seasonality, storage costs, and cleaner carry details.
```

12.3 Inter-commodity spreads

Inter-commodity spreads were more useful. They capture relative grain value, substitution effects, crop-specific demand, and wheat-class relative value.

```
Examples tested:
CORN vs SOYBEAN
CORN vs WHEAT_SRW
SOYBEAN vs WHEAT_SRW
WHEAT_SRW vs WHEAT_HRW

Vol-hedged spread unit:
spread_position = long commodity A - hedge_ratio * commodity B
hedge_ratio = lagged_60d_pnl_vol_A / lagged_60d_pnl_vol_B
hedge_ratio clipped to a reasonable range, e.g. 0.25 to 4.00

Best use:
80% outright book + 20% inter-commodity spread overlay

Decision:
Useful extension candidate, not replacement for core outright strategy.
```

12.4 Crush / crack spreads

For grains, the equivalent of a crack spread is the soybean crush spread. It was not traded in this dataset.

```
Soybean crush spread concept:
soybean_crush = soybean_meal + soybean_oil - soybeans

Why not traded here:
The dataset does not contain soybean meal and soybean oil futures prices.

How crush is used:
Cargill crush is used as an informational feature only, not as a traded spread leg.
```

13. Why the strategy works

Effect	How strategy captures it	Economic rationale
Momentum	mom_60 and price trend components	Supply/demand shocks persist and futures trend when the market reprices.
Short-term reversal	rev_5	Weather headlines, speculative flows, and liquidity shocks can overshoot.
Term structure / scarcity	curve_spread, curve_ratio, curve_change	Backwardation and curve tightening indicate nearby scarcity or strong demand.
Hedging pressure	COT producer/merchant and money-manager positioning	Commercial hedging and speculative crowding contain different information.
Physical stock-flow	public inventories, receipts, Cargill inventory and crush	Actual flows and stocks can lead or validate futures repricing.
Conviction filter	prediction dispersion edge filter	Avoids trading when the model has no clear relative view.

14. Why the strategy might not work

Weakness	Problem	Mitigation
Small universe	Only four commodities means limited cross-sectional breadth.	Keep model simple; avoid high-flexibility ML.
Policy shocks	Tariffs or trade-war headlines can dominate normal curve/COT/physical logic.	Document risk; avoid hard-coded historical event filters.
Sparse physical data	Public and Cargill physical features may be lagged, local, or incomplete.	Use strong Ridge regularization and separate overlay.
Static model risk	Static edge-filtered result may be optimistic because diagnostics informed improvements.	Use annual walk-forward as honesty check.
Directional limitation	Cross-sectional demeaning can short the weakest grain even if all grains are bullish.	Future extension: directional overlay.
Regime dependency	Best in shock/repricing regimes, weaker in low-dispersion periods.	Use edge and opportunity filters to reduce exposure.

15. Recommendation

Best interview answer:

```
I would recommend the static edge-filtered two-block Ridge as the candidate trading strategy, because it has the
best tradable performance and a clear economic rationale.

However, I would present the annual 20-day walk-forward Ridge as the robustness check, because it proves the idea
survives when coefficients are learned only from past data.

For production, I would start with the outright relative-value book, monitor regime-specific performance, and keep
the 80% outright / 20% inter-commodity spread overlay as an extension candidate after testing on the 2021-February
2026 holdout.
```

Use case	Recommended version	Why
Interview main story	Static edge-filtered two-block Ridge	Best performance and clean economic explanation.
Robustness / anti-overfit	Annual 20-day walk-forward Ridge	Uses only prior data at each retrain and remains positive.
Execution improvement	2-day skip-rebalance static candidate	Higher OOS Sharpe and lower turnover in static test; needs external holdout.
Diversification extension	80% outright + 20% inter-commodity spread overlay	Improves drawdown/diversification but reduces OOS PnL.
High-conviction version	Multi-condition opportunity filter	Trades less; useful if mandate prioritizes Sharpe and crisis avoidance over capacity.

16. One-page cheat sheet

Question	Answer
What is the strategy?	Cross-sectional long/short relative-value strategy in grain futures.
What does it trade?	Mainly outright futures. Inter-commodity spreads are an extension. Calendar and crush spreads are not final trades.
What is the alpha?	Momentum/reversal, curve tightness, COT positioning, physical stock-flow data, and Cargill inventory/crush information.
Why Ridge?	Simple, robust, interpretable, and safer with correlated features and a small dataset.
Why two blocks?	Core features are more stable; physical data is valuable but noisy and needs stronger shrinkage.
Why edge filter?	Avoids trading when the model has no clear cross-sectional winner/loser.
When does it work?	Weather shocks, export bans, geopolitics, COVID recovery, demand repricing.
When does it fail?	Low-dispersion abundant supply and policy-driven regimes like US-China trade war.
What would I recommend?	Static edge-filtered Ridge as candidate; annual walk-forward Ridge as proof/check.

One sentence summary: This is a relative-value grain futures strategy that uses price trend/reversal, curve tightness, COT positioning, and physical-flow data to rank corn, soybeans, and wheat; it works best in shock and repricing regimes, struggles in policy-driven or low-dispersion regimes, and should be traded mainly through outright futures with inter-commodity spreads as a possible overlay.